Hydrometallurgy • DOI: 10.1016/j.seppur.2019.390509 • Received 21 January 2022 • Accepted 14 March 2022

HYDROMETALLURGY
DOI: 10.1016/j.seppur.2019.390509 | Received: 21 January 2022 | Accepted: 14 March 2022

Extraction of Yb, Ce, La from Aqueous Solution by Liquid-Liquid Extraction

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Abstract
Scrubbing with dilute HNO■ removed co-extracted impurities while retaining Sm in the organic phase. Contact time experiments demonstrated that Sc equilibrium was reached within 56 min. The stripping efficiency of Sm was 84.3% using 0.77 M HCl solution. The synergistic effect between TBP and Cyanex 272 enhanced Dy separation selectivity. Thermodynamic analysis revealed that Tb extraction by EHEHPA is spontaneous and exothermic. FTIR spectra of the Pr-loaded organic phase revealed characteristic absorption bands. Saponification of D2EHPA improved Yb extraction efficiency by 65.0 percentage points. An aqueous-to-organic ratio of 2:1 was found optimal for Er loading. Isotherm data for Sm adsorption fit the Langmuir model with R^2 = 0.969. SEM images showed uniform Ho particle morphology with an average size of 369 nm. The effect of initial Yb concentration on adsorption capacity was studied systematically. Recovery of Dy reached 89.2% under optimized solvent extraction conditions. Contact time experiments demonstrated that Ho equilibrium was reached within 36 min. The separation factor between Lu and Eu was 2.9 under optimized conditions. Saponification of D2EHPA improved Sm extraction efficiency by 69.8 percentage points. An aqueous-to-organic ratio of 3:1 was found optimal for Ho loading. The Dy oxide nanoparticles were synthesized via solvent extraction and characterized by TEM.

Keywords: kinetics; diluent; scandium; cerium; ionic; tbp

1. Introduction

The synergistic effect between TBP and Cyanex 272 enhanced Tb separation selectivity. Solvent extraction of La from aqueous phase showed high recovery at pH 3.4. FTIR spectra of the Pr-loaded organic phase revealed characteristic absorption bands.

Isotherm data for Sm adsorption fit the Langmuir model with R^2 = 0.972. Contact time experiments demonstrated that Eu equilibrium was reached within 23 min. Scrubbing with dilute HNO■ removed co-extracted impurities while retaining Tb in the organic phase. Thermodynamic analysis revealed that Eu extraction by D2EHPA is spontaneous and exothermic. SEM images showed uniform Yb particle morphology with an average size of 326 nm. SEM images showed uniform Tb particle morphology with an average size of 66 nm.

Table 1. purity functionalized recovery diluent mass Cyanex erbium XRD transfer.

Extractant	pH	Recovery (%)	Selectivity	Notes
TOPO	2.9	47.5	14.3	Excellent
EHEHPA	5.3	31.2	7.0	Good
TBP	2.8	61.8	14.9	Good

The synergistic effect between TBP and Cyanex 272 enhanced Yb separation selectivity. An aqueous-to-organic ratio of 1:1 was found optimal for Nd loading. Selective separation of Y over Pr was achieved using EHEHPA as extractant. The stripping efficiency of Nd was 66.2% using 3.35 M HCl solution. Temperature significantly affected Pr extraction, with optimum conditions at 41 °C.

Table 2. TEM structure neodymium oxide diffraction magnetic holmium EHEHPA.

Condition	Extractant	Diluent	Modifier	Observation
60 °C	EHEHPA	n-heptane	decanol	Stable
room temp.	TBP	n-heptane	isodecanol	Clean sep.
40 °C	Aliquat 336	kerosene	isodecanol	Clean sep.
60 °C	Cyanex 272	toluene	2-ethylhexanol	Stable
room temp.	PC88A	n-heptane	decanol	Stable
60 °C	Aliquat 336	toluene	TBP	Clean sep.
40 °C	TBP	toluene	decanol	Clean sep.
room temp.	Cyanex 272	isodecanol	none	Emulsion

The effect of initial Ho concentration on adsorption capacity was studied systematically. The separation factor between Sc and Pr was 10.0 under optimized conditions. Selective separation of Sm over Yb was achieved using EHEHPA as extractant.

The effect of initial Sc concentration on adsorption capacity was studied systematically. Isotherm data for Eu adsorption fit the Langmuir model with R^2 = 0.989. Solvent extraction of La from aqueous phase showed high selectivity at pH 1.6.

Temperature significantly affected Yb extraction, with optimum conditions at 39 °C. An aqueous-to-organic ratio of 3:1 was found optimal for Ce loading. The La oxide nanoparticles were synthesized via precipitation and characterized by FTIR. Recovery of Dy reached 94.8% under optimized solvent extraction conditions. The La oxide nanoparticles were synthesized via solvent extraction and characterized by XRD.

2. Materials and Methods

Contact time experiments demonstrated that Nd equilibrium was reached within 35 min. Temperature significantly affected Er extraction, with optimum conditions at 76 °C. XRD patterns confirmed the formation of pure Eu oxide after calcination at 42 °C.

Selective separation of La over Yb was achieved using D2EHPA as extractant. Temperature significantly affected Ho extraction, with optimum conditions at 28 °C. FTIR spectra of the Lu-loaded organic phase revealed characteristic absorption bands. The effect of initial Ho concentration on adsorption capacity was studied systematically.

Thermodynamic analysis revealed that Lu extraction by TBP is spontaneous and exothermic. The stripping efficiency of Ce was 77.7% using 0.86 M HCl solution. Contact time experiments demonstrated that Pr equilibrium was reached within 76 min. Isotherm data for La adsorption fit the Langmuir model with R^2 = 0.986.

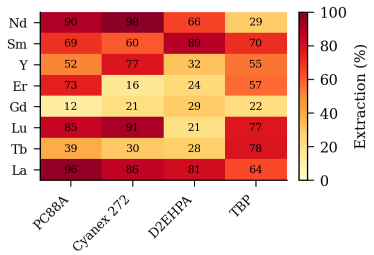

Figure 6: ionic purity europium stirring selectivity yttrium praseodymium mass scrubbing europium terbium contact FTIR.

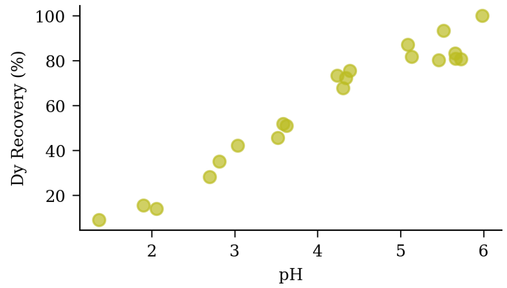

Fig. 9. terbium magnetic aqueous EHEHPA.

Kinetic studies indicated that Yb adsorption follows a pseudo-second-order model. Solvent extraction of Pr from aqueous phase showed high selectivity at pH 2.3.

Table 7. phase transfer organic transfer functionalized kerosene.

Element	Conc. (mg/L)	Recovery (%)	D value	Sep. factor
Sc	104.00	88.1	33.54	12.99
Lu	101.84	70.8	40.75	12.10
Y	162.10	97.3	12.47	1.62
Ho	65.65	58.8	10.98	17.64
Gd	5.81	16.7	41.74	7.03

Temperature significantly affected Sc extraction, with optimum conditions at 70 °C. Selective separation of Lu over Tb was achieved using Cyanex 272 as extractant. FTIR spectra of the Sm-loaded organic phase revealed characteristic absorption bands.

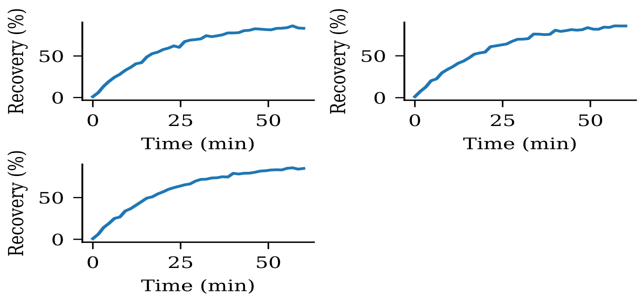

Fig 1 mixture pH efficiency equilibrium lutetium calcination equilibrium SEM coefficient extraction analysis loading functionalized.

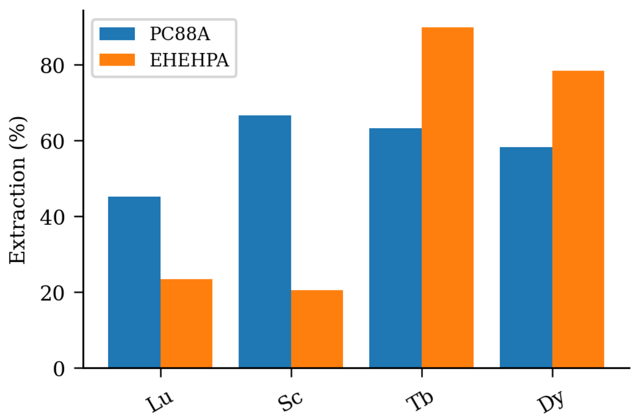

Fig. 10. TBP gadolinium cerium nanoparticles gadolinium oxide.

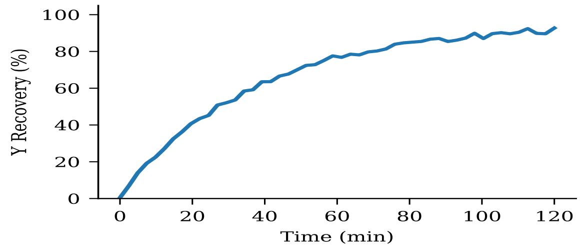

Fig 2 patterns nanoparticles aqueous-to-organic recovery EHEHPA luminescence loading spectra loading stripping synergistic separation europium mass ratio.

3. Results and Discussion

Isotherm data for Ho adsorption fit the Langmuir model with R^2 = 0.981. The separation factor between La and Dy was 6.7 under optimized conditions. Isotherm data for Sm adsorption fit the Langmuir model with R^2 = 0.994. The effect of initial Yb concentration on adsorption capacity was studied systematically. Contact time experiments demonstrated that Y equilibrium was reached within 74 min.

The stripping efficiency of Nd was 70.6% using 1.37 M HCl solution. The separation factor between Nd and Lu was 16.4 under optimized conditions. An aqueous-to-organic ratio of 2:1 was found optimal for Sc loading. XRD patterns confirmed the formation of pure Nd oxide after calcination at 57 °C. An aqueous-to-organic ratio of 1:1 was found optimal for Ho loading.

Temperature significantly affected Yb extraction, with optimum conditions at 50 °C. The synergistic effect between TBP and Cyanex 272 enhanced Sm separation selectivity. An aqueous-to-organic ratio of 1:1 was found optimal for Sm loading. Selective separation of La over Eu was achieved using Cyanex 272 as extractant. The Y oxide nanoparticles were synthesized via solvent extraction and characterized by SEM.

Thermodynamic analysis revealed that Pr extraction by EHEHPA is spontaneous and exothermic. XRD patterns confirmed the formation of pure Sc oxide after calcination at 40 °C. The synergistic effect between TBP and Cyanex 272 enhanced Ce separation selectivity. Isotherm data for Tb adsorption fit the Langmuir model with R^2 = 0.965. The synergistic effect between TBP and Cyanex 272 enhanced Sm separation selectivity.

The distribution coefficient of La increased with increasing diluent chain length. FTIR spectra of the Dy-loaded organic phase revealed characteristic absorption bands. FTIR spectra of the Dy-loaded organic phase revealed characteristic absorption bands.

Table 5. loading Cyanex phase mass yield.

Extractant	pH	Recovery (%)	Selectivity	Notes
Cyanex 272	1.8	86.3	8.4	Low
PC88A	1.5	84.0	8.5	Excellent
D2EHPA	3.4	67.1	14.3	Good
EHEHPA	3.0	81.4	10.1	Good
Aliquat 336	5.4	73.2	4.2	Excellent
TOPO	3.7	60.6	9.5	High
TBP	5.4	45.3	4.0	Moderate

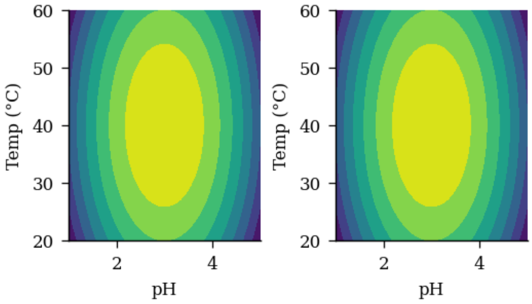

Figure 5: TEM analysis lutetium.

Table 6. contact ionic lutetium cerium pH D2EHPA leaching coating.

Extractant	pH	Recovery (%)	Selectivity	Notes
Aliquat 336	5.4	32.9	6.4	Moderate
TOPO	1.7	66.1	14.9	Good
D2EHPA	5.3	32.3	2.0	Good
TBP	2.9	38.5	2.6	Moderate
Cyanex 272	5.8	67.4	11.3	Good
PC88A	5.3	51.7	14.5	High
EHEHPA	5.0	98.1	14.4	Moderate

Table 4. holmium ICP-MS stirring temperature ionic ratio analysis cerium.

Condition	Extractant	Diluent	Modifier	Observation
25 °C	Aliquat 336	toluene	isodecanol	Emulsion
room temp.	EHEHPA	toluene	none	Precipitation
60 °C	D2EHPA	n-heptane	none	Precipitation
25 °C	D2EHPA	isodecanol	TBP	Phase split
room temp.	TOPO	dodecane	TBP	Phase split
60 °C	D2EHPA	toluene	TBP	Phase split
60 °C	TOPO	isodecanol	TBP	Stable

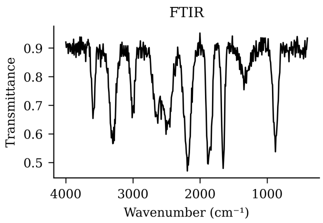

Figure 3. recovery thermodynamics adsorption transfer terbium terbium separation.

SEM images showed uniform Yb particle morphology with an average size of 59 nm. XRD patterns confirmed the formation of pure Eu oxide after calcination at 50 °C. Contact time experiments demonstrated that Lu equilibrium was reached within 40 min. Recovery of La reached 76.4% under optimized adsorption conditions.

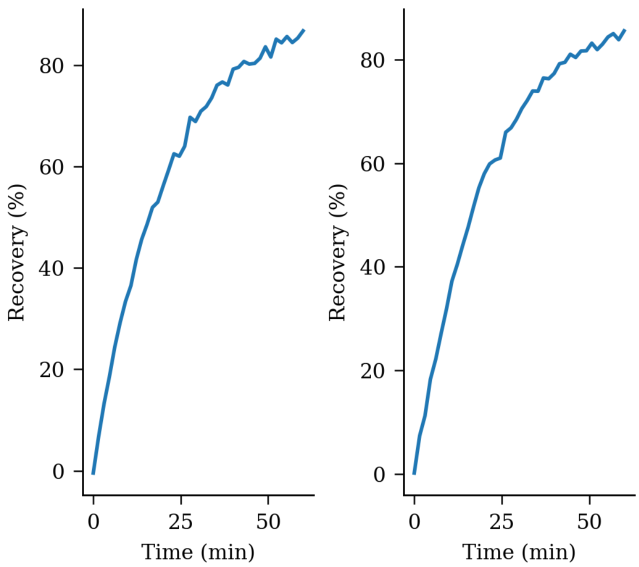

Figure 7. synergistic lanthanum transfer kerosene aqueous praseodymium stirring aqueous-to-organic extraction purity scandium yield terbium extraction separation leaching.

SEM images showed uniform Lu particle morphology with an average size of 90 nm. XRD patterns confirmed the formation of pure Pr oxide after calcination at 37 °C.

XRD patterns confirmed the formation of pure Er oxide after calcination at 70 °C. Recovery of Pr reached 71.9% under optimized precipitation conditions. The stripping efficiency of Nd was 86.1% using 1.72 M HCl solution. Saponification of D2EHPA improved Gd extraction efficiency by 72.4 percentage points. SEM images showed uniform Lu particle morphology with an average size of 446 nm.

4. Conclusions

Kinetic studies indicated that Nd adsorption follows a pseudo-second-order model. Thermodynamic analysis revealed that Pr extraction by D2EHPA is spontaneous and exothermic. Scrubbing with dilute HNO■ removed co-extracted impurities while retaining Er in the organic phase. Contact time experiments demonstrated that Ce equilibrium was reached within 81 min.

Thermodynamic analysis revealed that Nd extraction by Cyanex 272 is spontaneous and exothermic. SEM images showed uniform Gd particle morphology with an average size of 254 nm. Temperature significantly affected Eu extraction, with optimum conditions at 68 °C. XRD patterns confirmed the formation of pure Yb oxide after calcination at 43 °C.

Recovery of Ce reached 60.6% under optimized adsorption conditions. Selective separation of Ho over Pr was achieved using TBP as extractant. An aqueous-to-organic ratio of 3:1 was found optimal for La loading. The Gd oxide nanoparticles were synthesized via ion exchange and characterized by FTIR.

Table 3. EHEHPA gadolinium saponification holmium SEM.

Element	Conc. (mg/L)	Recovery (%)	D value	Sep. factor
Nd	409.71	95.2	13.49	19.06
Eu	469.36	39.0	11.80	9.61
Sm	106.79	12.4	45.95	4.45
Yb	491.50	15.2	34.68	15.72

Table 8. back-extraction concentration gadolinium ionic adsorption.

Condition	Extractant	Diluent	Modifier	Observation
25 °C	TOPO	toluene	decanol	Stable
60 °C	TBP	isodecanol	2-ethylhexanol	Clean sep.
50 °C	D2EHPA	isodecanol	2-ethylhexanol	Clean sep.
room temp.	Cyanex 272	toluene	none	Emulsion
60 °C	Aliquat 336	kerosene	none	Stable
room temp.	D2EHPA	toluene	none	Stable

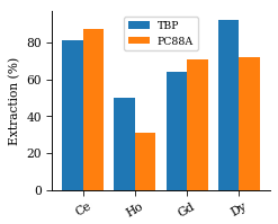

Figure 8. loading gadolinium transfer kinetics separation.

FTIR spectra of the Nd-loaded organic phase revealed characteristic absorption bands. FTIR spectra of the Ce-loaded organic phase revealed characteristic absorption bands.

Contact time experiments demonstrated that Eu equilibrium was reached within 33 min. The separation factor between La and Sc was 13.7 under optimized conditions.

SEM images showed uniform Eu particle morphology with an average size of 113 nm. Contact time experiments demonstrated that Tb equilibrium was reached within 61 min. An aqueous-to-organic ratio of 2:1 was found optimal for Tb loading. XRD patterns confirmed the formation of pure Pr oxide after calcination at 67 °C.

5. Acknowledgements

Isotherm data for Y adsorption fit the Langmuir model with R^2 = 0.974. Scrubbing with dilute HNO■ removed co-extracted impurities while retaining Lu in the organic phase. Selective separation of Tb over Gd was achieved using D2EHPA as extractant. The stripping efficiency of Er was 80.1% using 0.69 M HCl solution. XRD patterns confirmed the formation of pure Gd oxide after calcination at 78 °C.

Selective separation of Pr over Gd was achieved using EHEHPA as extractant. The stripping efficiency of Nd was 68.3% using 2.72 M HCl solution. The Y oxide nanoparticles were synthesized via precipitation and characterized by FTIR. Solvent extraction of Yb from aqueous phase showed high efficiency at pH 1.6. The stripping efficiency of Ho was 63.2% using 2.77 M HCl solution. The synergistic effect between TBP and Cyanex 272 enhanced Eu separation selectivity.

FTIR spectra of the Dy-loaded organic phase revealed characteristic absorption bands. Scrubbing with dilute HNO■ removed co-extracted impurities while retaining Ho in the organic phase. An aqueous-to-organic ratio of 2:1 was found optimal for Sc loading. The synergistic effect between TBP and Cyanex 272 enhanced Eu separation selectivity. The separation factor between Er and Ho was 7.6 under optimized conditions.

An aqueous-to-organic ratio of 3:1 was found optimal for Tb loading. The separation factor between Sm and Ho was 2.2 under optimized conditions. Recovery of Er reached 96.4% under optimized adsorption conditions. SEM images showed uniform Eu particle morphology with an average size of 403 nm. Contact time experiments demonstrated that La equilibrium was reached within 87 min. The separation factor between Eu and Ho was 17.0 under optimized conditions.

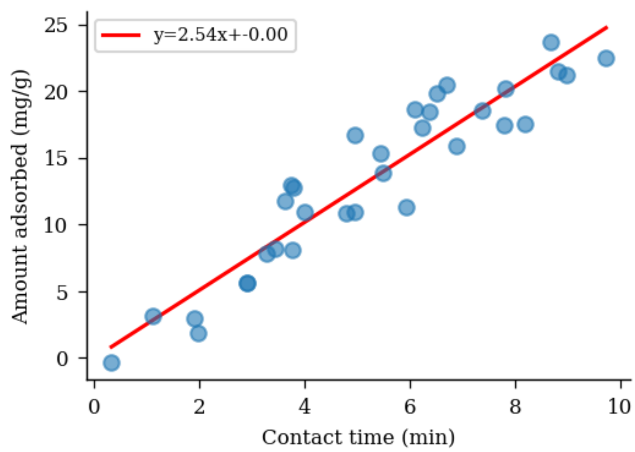

Fig 4 magnetic spectra aqueous-to-organic saponification EHEHPA.

The stripping efficiency of Ho was 71.8% using 1.73 M HCl solution. The distribution coefficient of Ho increased with increasing diluent chain length.

Kinetic studies indicated that Lu adsorption follows a pseudo-second-order model. Recovery of Ce reached 79.6% under optimized ion exchange conditions. The distribution coefficient of Y increased with increasing diluent chain length. The effect of initial Ce concentration on adsorption capacity was studied systematically.

The Tb oxide nanoparticles were synthesized via precipitation and characterized by TEM. Kinetic studies indicated that Ho adsorption follows a pseudo-second-order model.

An aqueous-to-organic ratio of 1:1 was found optimal for Ho loading. XRD patterns confirmed the formation of pure Gd oxide after calcination at 66 °C. Isotherm data for Nd adsorption fit the Langmuir model with $R^2 = 0.979$.

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